PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE C · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

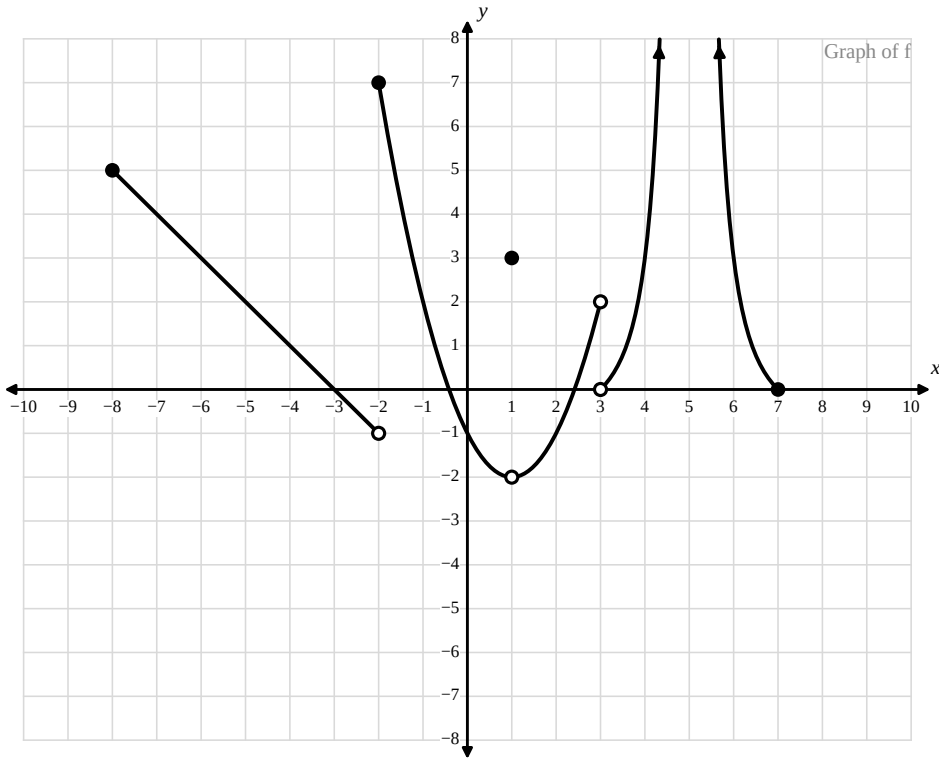
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

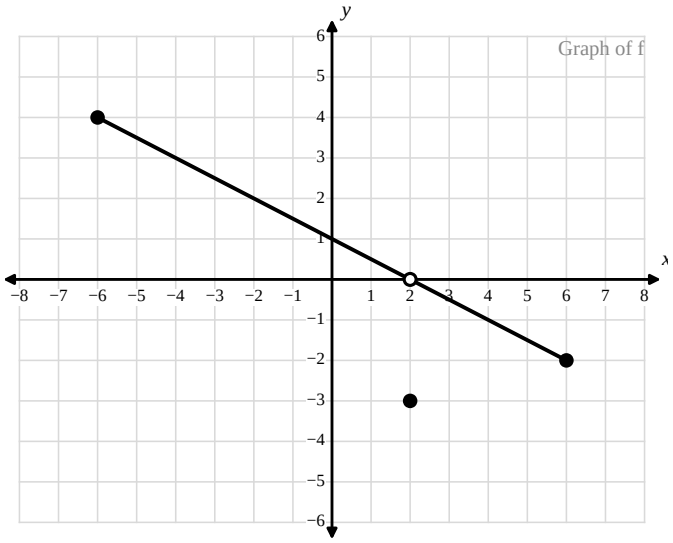


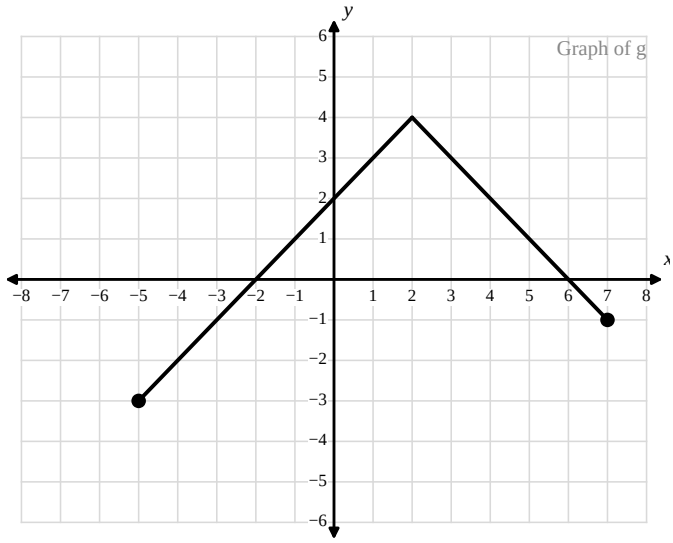
1. $\lim_{x\to 1} f(x) =$	**2.** $\lim_{x\to 3} f(x) =$	**3.** $\lim_{x\to 5} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = -1$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to 4} [f(x)\, g(x)]$

(2 points)

6. Use the table to estimate $\lim_{x \to 2} h(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	1.9	1.99	1.999	$\to 2 \leftarrow$	2.001	2.01	2.1
$h(x)$	11.41	11.9401	11.994001		12.006001	12.0601	12.61

$\lim_{x \to 2^-} h(x) \approx$ _____ $\lim_{x \to 2^+} h(x) \approx$ _____ $\lim_{x \to 2} h(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim_{x \to 1} f(x), \quad f(x) = \begin{cases} x^2 + 1, & x < 1 \\ 3 - x, & x \geq 1 \end{cases}$

8. $\lim_{x \to 3} \dfrac{2x^2 - 5x - 3}{x - 3}$

9. $\lim_{x \to 4} \dfrac{3 - \sqrt{x + 5}}{x - 4}$

10. $\lim_{x \to 4} \dfrac{\frac{1}{x} - \frac{1}{4}}{x - 4}$

11. Find the average rate of change of $f(x) = \sqrt{x}$ from 4 to x, then find its limit as $x \to 4$.

Each step: the work on the left, the reason on the right. "lim" stays on every line until the line where you substitute.

1. $\boxed{-2}$

Both sides of the parabola head to the open circle at the vertex $(1, -2)$.	The dot at $(1, 3)$ is $f(1)$; a limit ignores it.

2. $\boxed{\text{DNE}}$

$\lim\limits_{x \to 3^-} f(x) = 2,\ \lim\limits_{x \to 3^+} f(x) = 0$	Left: the parabola ends at the open circle $(3, 2)$. Right: the curve starts at the open circle $(3, 0)$.
$2 \neq 0$, so $\lim_{x \to 3} f(x)$ does not exist.	Sides disagree.

3. $\boxed{\infty}$

Both sides of $x = 5$ rise without bound.	Same direction from both sides.

4. $\boxed{a = 0,\ a = 2}$

Height -1 is reached from both sides at $x = 0$ and $x = 2$.	Two-sided approach.
Not $x = -2$: only the left line heads to -1; the right side starts at 7.	One-sided does not count.

5. $\boxed{-2}$

$\lim\limits_{x \to 4} [f(x)g(x)] = \left[\lim\limits_{x \to 4} f(x)\right]\left[\lim\limits_{x \to 4} g(x)\right]$	Limit of a product. Write the property first.
$= (-1)(2) = -2$	f is a line through $(4, -1)$; g is at height 2 on both sides of $x = 4$.

6. $\boxed{\approx 12}$

Left: 11.41, 11.9401, 11.994001	Rounds to 11.994, heading to 12.
Right: 12.61, 12.0601, 12.006001	Rounds to 12.006, heading to 12.
$\lim\limits_{x \to 2} h(x) \approx 12$	Both sides agree. An estimate, not a proof.

7. $\boxed{2}$

$\lim\limits_{x \to 1^-} f(x) = \lim\limits_{x \to 1^-} \left(x^2 + 1\right) = 1 + 1 = 2$	Left piece.
$\lim\limits_{x \to 1^+} f(x) = \lim\limits_{x \to 1^+} (3 - x) = 3 - 1 = 2$	Right piece.
$\lim\limits_{x \to 1} f(x) = 2$	$2 = 2$.

8. $\boxed{7}$

Substitution produces 0/0: indeterminate.	Factor.
$\lim\limits_{x \to 3} \dfrac{2x^2 - 5x - 3}{x - 3} = \lim\limits_{x \to 3} \dfrac{(2x + 1)(x - 3)}{x - 3}$	Trinomial: $2 \cdot (-3) = -6$; -6 and 1 add to -5.
$= \lim\limits_{x \to 3} (2x + 1) = 2(3) + 1 = 7$	Cancel $x - 3$; substitute.

9. $\boxed{-\frac{1}{6}}$

Substitution produces 0/0: indeterminate.	Multiply by the conjugate $3+\sqrt{x+5}$.
$\lim\limits_{x\to 4}\dfrac{3-\sqrt{x+5}}{x-4}=\lim\limits_{x\to 4}\dfrac{9-(x+5)}{(x-4)\left(3+\sqrt{x+5}\right)}$	Top is a^2-b^2.
$=\lim\limits_{x\to 4}\dfrac{-(x-4)}{(x-4)\left(3+\sqrt{x+5}\right)}$	$4-x=-(x-4)$. Watch the sign.
$=\lim\limits_{x\to 4}\dfrac{-1}{3+\sqrt{x+5}}=\dfrac{-1}{3+3}=-\dfrac{1}{6}$	Cancel; substitute.

10. $\boxed{-\frac{1}{16}}$

Substitution produces 0/0: indeterminate.	Multiply top and bottom by the LCD $4x$.
$\lim\limits_{x\to 4}\dfrac{\frac{1}{x}-\frac{1}{4}}{x-4}=\lim\limits_{x\to 4}\dfrac{4-x}{4x(x-4)}$	$4x\cdot\frac{1}{x}=4$, $4x\cdot\frac{1}{4}=x$.
$=\lim\limits_{x\to 4}\dfrac{-(x-4)}{4x(x-4)}=\lim\limits_{x\to 4}\dfrac{-1}{4x}$	Factor out -1; cancel.
$=\dfrac{-1}{4(4)}=-\dfrac{1}{16}$	Substitute.

11. $\boxed{\frac{\sqrt{x}-2}{x-4};\ \frac{1}{4}}$

$\dfrac{f(x)-f(4)}{x-4}=\dfrac{\sqrt{x}-2}{x-4}$	Average rate of change from 4 to x. This is the first answer.
$\lim\limits_{x\to 4}\dfrac{\sqrt{x}-2}{x-4}=\lim\limits_{x\to 4}\dfrac{\sqrt{x}-2}{(\sqrt{x}-2)(\sqrt{x}+2)}$	Substitution produces 0/0; $x-4$ is a difference of squares.
$=\lim\limits_{x\to 4}\dfrac{1}{\sqrt{x}+2}=\dfrac{1}{2+2}=\dfrac{1}{4}$	Cancel; substitute.